

Global Warming: Scientific Basis, Observed Impacts, Future Risks, and Evidence-Based Responses

Evidence Synthesis Report

May 2, 2026

Abstract

Global warming is the long-term heating of Earth’s surface, while climate change includes global warming plus broader changes in sea level, ice, precipitation, oceans, ecosystems, extremes, and human systems [2]. The assessed literature, especially the Intergovernmental Panel on Climate Change (IPCC) Sixth Assessment Report (AR6), supports the conclusion that human activities, principally greenhouse-gas emissions, have unequivocally caused global warming, with global surface temperature reaching about 1.1°C above 1850–1900 in 2011–2020 [1]. The main mechanism is the enhanced greenhouse effect: human-emitted greenhouse gases alter Earth’s radiative balance by absorbing and re-emitting outgoing infrared radiation [3, 4, 5]. This report synthesizes evidence on definitions, physical mechanisms, human drivers, observed impacts, future risks, mitigation, adaptation, climate justice, uncertainty, and misinformation. No independent empirical statistical test was performed because no empirical quantitative dataset was provided; a synthetic workflow demonstration is quarantined in an appendix and is not used as climate evidence.

Executive Summary

- **Definition.** Global warming is the long-term increase in Earth’s average surface temperature. Climate change is broader and includes warming plus shifts in precipitation, sea level, ice, oceans, extreme events, ecosystems, and human systems [2].
- **Main cause.** The assessed authoritative literature concludes that human activities, principally greenhouse-gas emissions, have unequivocally caused recent global warming [1, 7].

- **Physical mechanism.** Greenhouse gases absorb and re-emit outgoing infrared radiation, strengthening the natural greenhouse effect. Carbon dioxide (CO₂), methane (CH₄), nitrous oxide (N₂O), and fluorinated gases are major anthropogenic greenhouse gases [3, 4, 5].
- **Human drivers.** The major drivers include fossil fuel combustion, deforestation and land-use change, agriculture, industrial processes, and consumption patterns. EPA identifies fossil fuel use, deforestation, agriculture, waste, energy production, fertilizer use, refrigeration, and industrial processes as important sources of greenhouse gases [4].
- **Observed impacts.** Observed consequences include global temperature rise, greater warming over land than ocean, sea level rise, ocean heat uptake, ice loss, Arctic sea ice decline, changing extremes, ecosystem disruption, and rising risks to food, water, infrastructure, health, and economies [1, 6].
- **Future risks.** Risks increase with every additional increment of warming. Low-emissions pathways reduce projected warming and risk compared with high-emissions pathways, while adaptation options become more constrained as warming increases [1, 7].
- **Mitigation.** Evidence-supported mitigation portfolios include clean electricity, renewable energy, energy efficiency, electrification, methane reduction, avoided deforestation, sustainable agriculture, selected carbon capture for hard-to-abate sectors, and demand-side changes [1, 6].
- **Adaptation.** Adaptation options include heat action plans, flood defenses, resilient agriculture, water management, urban planning, disaster preparedness, ecosystem-based adaptation, and managed retreat where protection is not viable [1].
- **Equity.** Responsibility, vulnerability, and adaptive capacity are unevenly distributed. IPCC AR6 notes unequal historical and ongoing contributions from energy use, land use, lifestyles, and patterns of consumption and production [1].
- **Uncertainty and misinformation.** Uncertainty remains in areas such as regional precipitation, aerosol effects, cloud feedbacks, ice-sheet dynamics, tipping thresholds, technology deployment, and policy implementation. These uncertainties do not undermine the high-confidence conclusion that human activities are the dominant cause of recent global warming [1, 5].

Contents

1	Introduction	3
2	Definitions: Global Warming vs. Climate Change	3
2.1	Global warming	3
2.2	Climate change	4
3	Scientific Basis of Global Warming	4
3.1	The greenhouse effect	4
3.2	Radiative forcing	4
3.3	Major greenhouse gases	6
3.4	Carbon cycle feedbacks	6
4	Main Human Drivers of Global Warming	7
4.1	Fossil fuel combustion	7
4.2	Deforestation and land-use change	7
4.3	Agriculture	7
4.4	Industrial processes	7
4.5	Consumption patterns	8
5	Observed Global and Regional Trends	9
5.1	Global temperature trends	9
5.2	Sea level rise	9
5.3	Ocean heat content	9
5.4	Glacier and ice sheet loss	9
5.5	Arctic sea ice decline	10
5.6	Extreme weather patterns	10
5.7	Ecosystem changes	10
5.8	Human health impacts	10
5.9	Food, water, infrastructure, and economic impacts	10
6	Scientific Confidence and Attribution	11
7	Future Warming Scenarios and Risk Thresholds	12
7.1	Emissions pathways	12
7.2	The 1.5°C and 2°C thresholds	13
7.3	Long-term risks	13

8	Mitigation Options	14
8.1	Renewable energy	14
8.2	Energy efficiency	14
8.3	Electrification	14
8.4	Nuclear power	15
8.5	Carbon pricing	15
8.6	Methane reduction	15
8.7	Reforestation and avoided deforestation	15
8.8	Sustainable agriculture	15
8.9	Carbon capture	15
8.10	Behavioral and demand-side changes	16
9	Adaptation Options	17
9.1	Heat action plans	17
9.2	Flood defenses	17
9.3	Resilient agriculture	17
9.4	Water management	17
9.5	Urban planning	17
9.6	Disaster preparedness	18
9.7	Ecosystem-based adaptation	18
9.8	Managed retreat	18
10	Climate Justice and Equity	18
11	Major Uncertainties and Contested Areas	20
12	Common Misinformation Claims	20
13	Practical Policy and Action Synthesis	23
14	Limitations and Data Provenance	23
15	Conclusion	25
A	Synthetic Workflow Demonstration	25
B	References	25

1 Introduction

Global warming is scientifically important because it reflects a persistent imbalance in Earth’s energy budget. It is socially, economically, and politically important because warming alters water availability, coastal risk, extreme heat, ecosystems, food production, infrastructure planning, public health, and development choices. The purpose of this report is to provide a comprehensive evidence synthesis explaining what global warming is, why it is occurring, how confident scientists are about its causes and consequences, and what practical mitigation and adaptation actions are most supported by the supplied evidence.

This report distinguishes among four categories of evidence. First, physical science findings explain the mechanisms and attribution of warming. Second, impact and risk assessments evaluate observed and projected consequences. Third, mitigation evidence evaluates options to reduce greenhouse-gas emissions or enhance removals. Fourth, adaptation evidence evaluates options to reduce harm from climate hazards. Ethical, economic, and justice considerations cut across all four categories.

The report also distinguishes assessed literature from independent empirical testing. The research hypothesis supplied for this project states that cumulative anthropogenic greenhouse-gas emissions have been the dominant causal driver of observed global mean surface warming since 1850 and that low-emissions pathways produce lower future warming and climate-risk indicators than medium- or high-emissions pathways. The assessed authoritative literature supports this conclusion, especially IPCC AR6 [1, 6, 7]. However, no independent empirical hypothesis test was performed in this report because no empirical quantitative dataset was provided.

2 Definitions: Global Warming vs. Climate Change

2.1 Global warming

Global warming refers specifically to the long-term increase in Earth’s average surface temperature. NASA describes global warming as the long-term warming of the planet and notes that global temperature has shown a well-documented rise, especially since the late twentieth century [2]. In contemporary scientific and policy usage, global warming usually refers to warming driven primarily by human activities, especially fossil fuel combustion and associated greenhouse-gas emissions [1, 2].

2.2 Climate change

Climate change is broader than global warming. It includes changes in temperature, precipitation, sea level, ice cover, ocean conditions, extreme weather patterns, ecosystems, and regional or seasonal climate behavior. NASA states that climate change encompasses global warming but also includes rising sea levels, shrinking mountain glaciers, accelerating ice melt, and biological timing shifts [2]. Thus, global warming is one major component of climate change, but climate change includes the wider physical, ecological, and human-system consequences of warming.

3 Scientific Basis of Global Warming

3.1 The greenhouse effect

The greenhouse effect is a natural process that helps keep Earth habitable. Solar radiation enters the climate system; some is reflected, and some is absorbed by Earth's surface. The warmed surface emits infrared radiation. Greenhouse gases absorb and re-emit part of that outgoing infrared radiation, including back toward the surface. This natural greenhouse effect is necessary for a habitable climate.

Global warming is caused by an enhanced greenhouse effect. Human activities add long-lived greenhouse gases to the atmosphere, increasing the amount of outgoing infrared energy retained in the climate system. NOAA describes CO₂ as Earth's most important long-lived greenhouse gas and states that adding more CO₂ amplifies the natural greenhouse effect, causing global temperature to rise [3]. The Royal Society and U.S. National Academy of Sciences likewise explain the evidence and causes of climate change in terms of greenhouse-gas physics and converging observational evidence [5].

3.2 Radiative forcing

Radiative forcing is a change in Earth's energy balance caused by a forcing agent. Positive forcing tends to warm the climate system; negative forcing tends to cool it. Greenhouse gases exert positive forcing. Some aerosols cool by reflecting sunlight, while others, such as black carbon, can warm regionally. Land-use changes can affect albedo and carbon storage. Solar and volcanic influences affect climate, especially on shorter timescales, but IPCC AR6 attributes recent long-term global warming primarily to human influence [1].

Table 1: Major forcing agents and expected climate effects

Forcing agent	Main source	Direction of effect	Confidence	Notes
Carbon dioxide	Fossil fuels, cement production, deforestation, land-use change	Warming	Very high	Dominant long-lived anthropogenic greenhouse gas
Methane	Agriculture, fossil fuel systems, waste	Warming	High	Strong near-term warming influence
Nitrous oxide	Fertilizer, manure, industry, combustion	Warming	High	Long-lived greenhouse gas
Fluorinated gases	Refrigeration, air conditioning, industrial uses	Warming	High	Often high warming effect per molecule
Aerosols	Combustion, industry, biomass burning	Mixed, often cooling	Medium to high	Effects vary by composition and region
Solar and volcanic factors	Natural processes	Variable	High for short-term effects	Do not explain the observed long-term recent warming trend in IPCC assessment

3.3 Major greenhouse gases

Carbon dioxide. CO₂ enters the atmosphere through burning fossil fuels, cement production, deforestation, land clearing, and other land-use changes. NOAA reports that natural sinks, including plant growth and ocean absorption, remove about half of human-emitted CO₂, while the rest remains in the atmosphere, so atmospheric CO₂ rises when emissions exceed sink uptake [3]. IPCC AR6 reports that atmospheric CO₂ concentration in 2019 was higher than at any time in at least 2 million years [1].

Methane. Methane is emitted from agriculture, waste, and energy systems. EPA identifies agricultural activities, waste management, energy production and use, and biomass burning as contributors to CH₄ emissions [4]. Because methane is potent over shorter timescales, methane reduction can help slow near-term warming, although CO₂ remains central to cumulative long-term warming.

Nitrous oxide. Nitrous oxide emissions are linked strongly to agriculture, especially fertilizer use, and also to chemical production and fossil fuel combustion [4]. IPCC AR6 reports that methane and nitrous oxide concentrations in 2019 were higher than at any time in at least 800,000 years [1].

Fluorinated gases. Fluorinated gases are synthetic gases used in refrigeration, air conditioning, and industrial processes. They are emitted in smaller total quantities than CO₂ but can have high warming effects per molecule [4].

3.4 Carbon cycle feedbacks

The carbon cycle includes natural sinks and sources in oceans, forests, soils, vegetation, and geological systems. Oceans and terrestrial ecosystems currently remove a substantial fraction of human-emitted CO₂ [3]. However, warming can weaken some sinks or create additional emissions through wildfire, forest stress, soil carbon loss, and permafrost thaw. The magnitude and timing of these feedbacks remain uncertain, but their direction matters because weakened sinks or additional carbon release would make stabilization more difficult.

Ocean uptake of CO₂ also causes ocean acidification. NOAA explains that CO₂ dissolves into ocean water, reacts to form carbonic acid, and lowers ocean pH, affecting marine organisms that build shells and skeletons [3]. Thus, CO₂ emissions produce both warming and chemical changes in the ocean.

4 Main Human Drivers of Global Warming

4.1 Fossil fuel combustion

Fossil fuel combustion is the central driver of anthropogenic CO₂ emissions. Coal, oil, and natural gas are used in electricity generation, transport, industry, heating, and buildings. EPA identifies fossil fuel use as the primary source of CO₂ and notes that CO₂ can also be emitted from landscapes through deforestation, land clearance, and soil degradation [4]. Because CO₂ accumulates, cumulative emissions are especially important for long-term warming.

4.2 Deforestation and land-use change

Deforestation and land-use change contribute to global warming by releasing stored carbon and reducing future carbon uptake. Forest clearing for agriculture, urbanization, or resource extraction removes carbon sinks and can also alter biodiversity, hydrology, and local climate. Restoration and improved land management can remove CO₂, but such approaches require attention to permanence, biodiversity, land tenure, and social rights [1, 4].

4.3 Agriculture

Agriculture emits methane from livestock and rice production, nitrous oxide from fertilizers and manure, and CO₂ from land clearing and soil carbon loss. EPA identifies agriculture as a contributor to methane and the primary source of nitrous oxide emissions [4]. Mitigation options include improving fertilizer efficiency, reducing food waste, changing livestock management, protecting soils, and reducing land clearing.

4.4 Industrial processes

Industrial processes emit greenhouse gases through fossil energy use and chemical reactions. Cement production releases CO₂ chemically as well as through fuel use. Steel, chemicals, and heavy industry are energy-intensive and can be difficult to decarbonize. Fluorinated gases arise from refrigeration, air conditioning, and industrial applications [4]. Hard-to-abate industrial emissions are one area where carbon capture may have a role, but it cannot substitute for broad emissions reductions.

4.5 Consumption patterns

Consumption patterns shape emissions through energy demand, material use, diets, transport choices, urban design, housing, and supply chains. Production-based emissions accounting attributes emissions to where goods are produced; consumption-based accounting attributes emissions to where goods are consumed. These approaches produce different responsibility profiles and are relevant to climate justice. IPCC AR6 notes unequal historical and ongoing contributions from energy use, land use, lifestyles, and patterns of consumption and production [1].

Table 2: Human drivers and mitigation levers

Human driver	Main gases	Main sectors	Evidence strength	Mitigation levers
Fossil fuels	CO ₂ , CH ₄	Power, transport, buildings, industry	Very high	Clean electricity, efficiency, electrification, demand reduction
Deforestation and land-use change	CO ₂	Land use, agriculture, urban expansion	High	Avoided deforestation, restoration, soil and forest protection
Agriculture	CH ₄ , CO ₂	N ₂ O, Livestock, rice, fertilizer, soils	High	Methane reduction, fertilizer efficiency, food waste reduction
Industry	CO ₂ , F-gases, N ₂ O	Cement, steel, chemicals, refrigeration	High	Process innovation, electrification, selected carbon capture
Consumption patterns	Multiple	Transport, food, housing, materials	High	Low-carbon choices, circular economy, urban redesign

5 Observed Global and Regional Trends

5.1 Global temperature trends

IPCC AR6 reports that global surface temperature was 1.09°C higher in 2011–2020 than in 1850–1900, with larger increases over land than over the ocean [1]. NASA similarly states that global warming is a long-term warming of the planet and that global temperatures have risen markedly since the late twentieth century [2]. Regional warming differs: land warms faster than ocean, high-latitude regions are especially sensitive, and local exposure depends on geography, infrastructure, and vulnerability.

5.2 Sea level rise

Sea level rise results from thermal expansion of warming seawater and added water from melting glaciers and ice sheets. The IPCC synthesis reports that deep, rapid, and sustained emissions reductions would limit further acceleration of sea level rise, while higher-emissions pathways produce larger sea level commitments [1]. Sea level rise varies regionally because of ocean circulation, land motion, gravitational effects of ice loss, and local coastal processes.

5.3 Ocean heat content

Ocean heat content has increased as the ocean absorbs heat from the planetary energy imbalance. Increasing ocean heat contributes to thermal expansion, sea level rise, marine heatwaves, and stress on marine ecosystems. IPCC AR6 synthesizes evidence of widespread changes in the ocean and broader climate system [1, 6]. NOAA’s discussion of atmospheric CO₂ also emphasizes the importance of ocean uptake for both carbon storage and ocean acidification [3].

5.4 Glacier and ice sheet loss

Mountain glaciers and the Greenland and Antarctic ice sheets contribute to sea level rise when they lose mass. NASA’s definition page identifies shrinking mountain glaciers and accelerating ice melt in Greenland, Antarctica, and the Arctic as part of the broader climate-change evidence associated with warming [2]. IPCC AR6 also identifies cryosphere change as part of the observed climate-system response [1, 6].

5.5 Arctic sea ice decline

Arctic sea ice decline affects albedo, ecosystems, Indigenous and local communities, marine access, and regional feedbacks. Reduced sea ice exposes darker ocean water, which absorbs more solar energy than ice, amplifying regional warming. NASA includes Arctic ice melt among the broader changes associated with climate change [2]. The exact timing of future summer ice-free conditions depends on future emissions and internal variability.

5.6 Extreme weather patterns

Evidence has strengthened for human influence on some extremes, particularly heat extremes and heavy precipitation [1]. Changes in drought, wildfire weather, tropical cyclones, floods, and compound events vary by region and event type. It is important to distinguish the physical trend in a type of hazard from the total disaster impact, which also depends on exposure, vulnerability, infrastructure, land management, and preparedness.

5.7 Ecosystem changes

Ecosystems respond to warming through species range shifts, phenological changes, coral bleaching, forest stress, and altered biodiversity risk. Ocean acidification caused by CO₂ uptake affects marine organisms that depend on carbonate chemistry [3]. IPCC AR6 warns that risks of species extinction and irreversible biodiversity loss increase with additional warming [1].

5.8 Human health impacts

The supplied source set does not include a dedicated WHO health reference, so health impacts are discussed only at the level supported by the IPCC synthesis. IPCC AR6 identifies widespread impacts and risks to human systems, including risks that interact with food, water, livelihoods, infrastructure, and health [1, 6]. Heat exposure is one of the clearest health-relevant pathways. Vulnerability is higher among older people, children, outdoor workers, low-income groups, and communities with limited adaptive capacity.

5.9 Food, water, infrastructure, and economic impacts

Climate risks affect agriculture, fisheries, water availability, flood and drought exposure, infrastructure reliability, labor productivity, disaster losses, and economic stability. These impacts are shaped by both hazards and vulnerability. IPCC AR6 emphasizes that climate

risks are widespread and that adaptation and mitigation choices affect the scale of future impacts [1, 6].

Table 3: Observed trend synthesis

Impact area	Observed trend	Main supplied sources	Regional differences	Confidence
Temperature	Increasing globally; land warming exceeds ocean warming	IPCC, NASA	Strong regional variation	Very high
Sea level	Rising from warming oceans and land ice loss	IPCC, NASA	Varies by coast and land motion	Very high
Ocean heat	Increasing as part of planetary energy imbalance	IPCC, NOAA	Ocean-basin differences	Very high
Ice and snow	Glacier retreat and ice-sheet mass loss contribute to sea level	IPCC, NASA	Strong cryosphere regionality	High
Arctic sea ice	Declining as part of Arctic climate change	IPCC, NASA	Arctic amplification important	High
Extremes	Heat and heavy precipitation evidence strongest; other extremes more regional	IPCC	Event-type dependent	Medium to high
Human systems	Rising food, water, health, infrastructure, and livelihood risks	IPCC	Exposure and vulnerability dominate outcomes	Medium to high

6 Scientific Confidence and Attribution

IPCC AR6 states that human activities, principally greenhouse-gas emissions, have unequivocally caused global warming [1]. This high confidence does not mean absolute certainty in every regional projection or every impact estimate. Rather, it reflects converging evidence from observations, physical understanding, radiative forcing calculations, paleoclimate evi-

Figure description.

A final empirical version would show observed global mean surface temperature anomalies relative to 1850–1900 or another stated baseline, using authoritative observational datasets. No empirical plot is generated here because no dataset was supplied.

Figure 1: Planned figure: global temperature trend.

Figure description.

A final empirical version would show atmospheric CO₂, CH₄, N₂O, and selected fluorinated-gas concentration trends, using NOAA or comparable monitoring data. No empirical plot is generated here because no dataset was supplied.

Figure 2: Planned figure: greenhouse-gas concentration trends.

dence, and model-based attribution.

Key attribution evidence includes observed increases in well-mixed greenhouse-gas concentrations, the radiative properties of greenhouse gases, surface and ocean warming, cryosphere changes, and comparisons of natural and human forcing. IPCC AR6 estimates that total human-caused warming from 1850–1900 to 2010–2019 was likely 0.8°C to 1.3°C, with greenhouse gases contributing warming and other human drivers, mainly aerosols, offsetting part of that warming [1]. Natural drivers and internal variability are not sufficient to explain the long-term warming trend assessed by IPCC [1, 5].

Scientific confidence is strongest at the global scale for the human cause of warming, the basic greenhouse mechanism, the warming trend, ocean heat uptake, and sea level rise. Confidence is lower for some regional projections, especially local precipitation changes, specific drought trends, future ice-sheet dynamics, and tipping thresholds.

7 Future Warming Scenarios and Risk Thresholds

7.1 Emissions pathways

Future warming depends on future greenhouse-gas emissions. Low-emissions pathways involve rapid reductions and eventual net zero CO₂, accompanied by strong reductions in methane and other greenhouse gases. Medium-emissions pathways involve slower or incomplete reductions. High-emissions pathways involve continued high fossil fuel use and greater cumulative emissions. IPCC AR6 emphasizes that future risks depend strongly on choices

made now and in the near term [1, 7].

Table 4: Emissions pathway categories

Pathway type	Approximate emissions trend	Expected climate outcome	Key assumptions	Risk level
Low emissions	Rapid decline toward net zero CO ₂	Lower warming and reduced risks	Strong mitigation and sustained policy implementation	Lower but not zero
Medium emissions	Slow decline, plateau, or partial mitigation	Continued warming and rising risks	Partial policy success and uneven transition	Moderate to high
High emissions	Continued growth or fossil-heavy development	Severe warming and larger impacts	Weak mitigation and high cumulative emissions	Very high

7.2 The 1.5°C and 2°C thresholds

The 1.5°C and 2°C thresholds refer to global warming relative to pre-industrial levels. They are policy-relevant because international climate goals use them as benchmarks, and they are scientifically important because risks increase with additional warming. IPCC AR6 reports that adaptation options feasible today become more constrained and less effective with increasing warming [1]. Overshoot pathways, in which warming temporarily exceeds a threshold and later declines, can still create additional and sometimes irreversible risks.

7.3 Long-term risks

Long-term risks include ice-sheet instability, long-term sea level rise, ecosystem transformation, coral reef loss, biodiversity loss, and limits to adaptation. IPCC AR6 states that the likelihood and impacts of abrupt or irreversible changes increase with further warming [1]. Some risks are high confidence in direction but uncertain in timing and magnitude. This is especially true for low-probability, high-impact outcomes such as rapid ice-sheet responses or tipping-element changes.

Figure description.

A final empirical version would show low-, medium-, and high-emissions projected warming pathways, clearly labeled as conditional projections rather than forecasts. No empirical scenario plot is generated here because no scenario dataset was supplied.

Figure 3: Planned figure: projected warming pathways.

8 Mitigation Options

Mitigation reduces greenhouse-gas emissions or enhances removals. IPCC AR6 synthesizes mitigation evidence across energy, land, industry, buildings, transport, demand-side measures, and carbon dioxide removal [6]. Because the supplied source set does not include detailed IPCC Working Group III tables, option ratings below are qualitative synthesis judgments and should not be read as quantitative cost or potential estimates.

8.1 Renewable energy

Solar, wind, hydropower, geothermal, and sustainable bioenergy can reduce fossil fuel combustion when deployed with appropriate grid, storage, transmission, and land-use planning. Solar and wind are especially important for clean electricity expansion. Hydropower can provide low-carbon electricity but has ecological and social tradeoffs. Bioenergy requires careful sustainability constraints.

8.2 Energy efficiency

Energy efficiency reduces the amount of energy needed to provide services in buildings, appliances, industry, and transport. Efficiency can reduce emissions, lower bills, and reduce infrastructure requirements. Rebound effects, upfront costs, and split incentives can limit effectiveness.

8.3 Electrification

Electrification shifts end uses such as vehicles, heating, and some industrial processes from direct fossil fuel combustion to electricity. It reduces emissions most effectively when electricity generation is increasingly low-carbon. Electrification requires grid expansion, charging infrastructure, supply-chain planning, and affordability policies.

8.4 Nuclear power

Nuclear power can provide low-carbon firm electricity in some national contexts. Its role depends on cost, construction timelines, safety governance, waste management, public acceptance, and compatibility with electricity-system planning. It should not be treated as either universally required or universally irrelevant.

8.5 Carbon pricing

Carbon taxes, cap-and-trade systems, and related pricing instruments can encourage emissions reductions by making greenhouse-gas emissions more costly. Effectiveness depends on price level, sectoral coverage, enforcement, complementary policies, and distributional design. Revenue recycling can reduce inequity.

8.6 Methane reduction

Methane reduction can slow near-term warming. Options include oil and gas leak detection and repair, coal mine methane controls, landfill gas capture, livestock management, manure management, and rice cultivation changes. EPA identifies agriculture, waste, energy production and use, and biomass burning as methane sources [4].

8.7 Reforestation and avoided deforestation

Avoided deforestation protects existing carbon stocks and biodiversity. Reforestation and ecosystem restoration can sequester carbon, improve water regulation, and support biodiversity. Key risks include impermanence, wildfire, drought stress, monoculture plantations, land conflict, and violations of Indigenous or local land rights.

8.8 Sustainable agriculture

Sustainable agriculture includes fertilizer efficiency, soil carbon protection, agroforestry, livestock management, food waste reduction, and dietary shifts where culturally appropriate and nutritionally adequate. These actions can reduce methane, nitrous oxide, and land-use pressure.

8.9 Carbon capture

Carbon capture and storage can be relevant for hard-to-abate industrial processes and possibly for some carbon dioxide removal pathways. Direct air capture and bioenergy with carbon

capture require substantial energy, infrastructure, monitoring, and storage governance. Carbon capture should not be used as a rationale for delaying emissions cuts.

8.10 Behavioral and demand-side changes

Demand-side mitigation includes reduced food waste, lower-emissions transport, efficient housing, material efficiency, circular economy approaches, and reduced high-emitting consumption. These options depend on infrastructure, affordability, social norms, policy design, and equity.

Table 5: Mitigation options: qualitative synthesis

Option	Emissions-reduction role	Readiness	Co-benefits	Key limits or risks
Solar and wind	High in power systems	High	Air quality, energy security	Grid integration, storage, permitting
Efficiency	High across sectors	High	Lower bills, reduced demand	Rebound effects, upfront costs
Electrification	High if powered by clean electricity	Medium to high	Air quality, reduced fossil use	Grid expansion and infrastructure
Methane reduction	High near-term relevance	High in many sources	Safety and air-quality benefits	Monitoring and enforcement
Forest protection	Carbon and biodiversity benefits	Medium to high	Water and ecosystem benefits	Permanence and land rights
Nuclear power	Context-dependent low-carbon electricity	Medium	Firm low-carbon power	Cost, time, waste, acceptance
Carbon capture	Sector-specific role	Medium to low	Hard-to-abate sectors	Cost, scale, storage governance
Demand-side changes	High in aggregate if enabled	Medium	Health and cost benefits	Equity and behavioral constraints

9 Adaptation Options

Adaptation reduces harm from climate impacts that are already occurring or unavoidable. IPCC AR6 states that feasible and effective adaptation options exist today but become constrained and less effective as warming increases [1]. Adaptation cannot substitute for mitigation, because higher warming increases residual risks and losses.

9.1 Heat action plans

Heat action plans include heat early warning systems, cooling centers, occupational protections, public health communication, urban shade, cool roofs, and targeted support for older people, children, outdoor workers, and medically vulnerable groups. Equity is central because heat exposure and access to cooling differ sharply within cities and regions.

9.2 Flood defenses

Flood defenses include seawalls, levees, drainage improvements, floodplain management, and nature-based buffers. They can reduce risk but may also create residual risk, high maintenance costs, or risk transfer to neighboring communities.

9.3 Resilient agriculture

Resilient agriculture includes drought-tolerant crops, crop diversification, shifting planting dates, soil moisture conservation, irrigation efficiency, and climate information services. Smallholder access, affordability, and local knowledge are crucial.

9.4 Water management

Water adaptation includes demand management, water reuse, groundwater protection, watershed restoration, drought planning, and cooperation over shared basins. Maladaptation risks include overreliance on groundwater or inequitable allocation during drought.

9.5 Urban planning

Urban planning can reduce heat and flood exposure through green infrastructure, cool roofs, zoning, transit-oriented development, floodplain management, building codes, and resilient design. Cities are important because they concentrate people, infrastructure, and heat exposure.

9.6 Disaster preparedness

Preparedness includes early warning systems, evacuation planning, emergency response, risk mapping, insurance and financial protection, and community-based disaster planning. Effectiveness depends on governance capacity and whether warnings reach vulnerable groups.

9.7 Ecosystem-based adaptation

Ecosystem-based adaptation uses mangroves, wetlands, forests, reefs, and watersheds to buffer hazards and support biodiversity. It must account for ecosystem limits under warming and protect local and Indigenous rights.

9.8 Managed retreat

Managed retreat may be necessary where chronic flooding, erosion, sea level rise, or repeated disasters make protection impractical. It raises difficult questions of consent, compensation, cultural loss, legal rights, and community continuity.

10 Climate Justice and Equity

Climate justice addresses unequal responsibility, unequal vulnerability, and unequal capacity to respond. IPCC AR6 reports unequal historical and ongoing contributions to greenhouse-gas emissions across regions, countries, and individuals [1]. Historical cumulative emissions, current annual emissions, per-capita emissions, and consumption-based emissions each illuminate different dimensions of responsibility.

Vulnerability differs by income, geography, governance, infrastructure, health systems, historical marginalization, and dependence on climate-sensitive livelihoods. Low-income countries, small island states, Indigenous communities, children, outdoor workers, elderly people, and future generations often face heightened risks or reduced adaptive capacity.

Finance, technology transfer, capacity building, and loss-and-damage arrangements are important because many vulnerable communities have contributed least to historical emissions but face significant adaptation needs. Procedural justice requires participation in decision-making, respect for land rights, and free, prior, and informed consent for Indigenous peoples where climate actions affect their territories or resources.

Table 6: Adaptation options: qualitative synthesis

Option	Main risk addressed	Evidence of usefulness	Equity considerations	Limits or risks
Heat action plans	Extreme heat	Strong in principle	Protect vulnerable groups	Outreach and funding
Flood defenses	Coastal and river flooding	Context-specific	Can shift risk	Cost, overtopping, maintenance
Resilient agriculture	Food insecurity	Medium to strong	Smallholder access	Limits under severe warming
Water management	Drought and scarcity	Strong in principle	Allocation conflicts	Groundwater depletion
Urban planning	Heat and flooding	Strong when integrated	Housing affordability	Governance capacity
Disaster preparedness	Extremes and compound hazards	Strong in principle	Inclusive warnings	Institutional capacity
Ecosystem-based adaptation	Floods, heat, coastal hazards	Context-specific	Land rights and consent	Ecosystem degradation
Managed retreat	Chronic exposure	Context-dependent	Consent and compensation	Cultural and social disruption

Table 7: Climate justice dimensions

Group or region		Main climate risks	Relative emissions responsibility	Key justice issues
Small island states		Sea level rise, storms, coastal loss	Low	Survival, relocation, loss and damage
Low-income countries		Heat, food, water, disasters	Low historical emissions	Finance, capacity, development rights
Indigenous communities	com-	Ecosystem disruption, land loss	Generally low	Land rights, sovereignty, consent
Future generations	genera-	Long-term warming and sea level rise	None	Intergenerational responsibility
Outdoor workers and elderly people	workers and elderly peo-	Heat exposure and health risk	Varies	Health protection and labor safeguards

11 Major Uncertainties and Contested Areas

The central scientific consensus is strong: human activities are the dominant cause of recent global warming [1, 5]. Remaining uncertainties concern magnitudes, regional details, thresholds, and implementation feasibility. Uncertainty does not imply safety; it can include the possibility of worse outcomes.

Scientific uncertainty is different from political disagreement. For example, uncertainty about the exact magnitude of future regional rainfall change does not undermine the high-confidence conclusion that greenhouse-gas emissions are warming the planet. Policy disagreements often concern values, costs, equity, feasibility, and responsibility rather than the basic physics.

12 Common Misinformation Claims

Table 8: Uncertainty categories

Topic	Well established	What remains uncertain	Why it matters
Human cause of warming	Human influence is dominant	Exact partitioning among forcing agents	Attribution and policy design
Climate sensitivity	Warming response is constrained by evidence	Remaining range of response	Carbon budgets and long-term risk
Aerosols and clouds	Affect radiative balance	Regional and feedback details	Near-term warming and air-quality tradeoffs
Regional rainfall	Hydrological patterns are changing	Local magnitude and timing	Water and agriculture planning
Sea level rise	Sea level is rising	Long-term ice-sheet response	Coastal planning and retreat
Tipping elements	Risks increase with warming	Thresholds and timing	Low-probability, high-impact risk
Technology deployment	Many mitigation options exist	Scale, speed, cost, governance	Feasibility of pathways
Adaptation limits	Adaptation becomes constrained at higher warming	Local thresholds and capacity	Residual losses and damages

Table 9: Common misinformation claims and evidence-based responses

Claim	Evidence-based response
“Climate has always changed, so humans are not responsible.”	Natural climate variability exists, but IPCC AR6 concludes that human activities, principally greenhouse-gas emissions, have unequivocally caused recent global warming [1].
“Scientists are not confident about global warming.”	Multiple independent lines of evidence support high confidence in human-caused warming, including observations, radiative physics, attribution studies, and paleoclimate evidence [1, 5].
“Cold weather disproves global warming.”	Weather is short-term; climate describes long-term statistical patterns. Global warming does not eliminate cold events [2].
“CO ₂ is too small a fraction of the atmosphere to matter.”	Trace gases can strongly affect radiative balance because they absorb infrared radiation. NOAA identifies CO ₂ as Earth’s most important long-lived greenhouse gas [3].
“Warming is caused by the Sun.”	IPCC attribution separates natural drivers from anthropogenic drivers and finds human influence dominant for recent global warming [1].
“Climate models are unreliable.”	Models have limitations, but IPCC assessments integrate model evidence with observations, physical understanding, and paleoclimate evidence rather than relying on any single model [6].
“Adaptation alone is enough.”	Adaptation is necessary, but IPCC states adaptation options become constrained and less effective with increasing warming [1].
“Carbon capture means emissions cuts are unnecessary.”	Carbon capture may help in selected hard-to-abate sectors, but assessed mitigation portfolios still require deep emissions reductions [1, 6].

13 Practical Policy and Action Synthesis

Evidence-supported climate action requires both mitigation and adaptation. Mitigation reduces future warming; adaptation reduces current and near-term harm. Equity and governance determine whether actions are effective and fair.

At the individual and household level, useful actions include energy efficiency, electrification where feasible, lower-carbon transport, reduced food waste, lower-emissions diets where appropriate, and civic engagement. These actions are most effective when supported by infrastructure and policy.

At the local and city level, priorities include urban heat planning, public transit, building codes, green infrastructure, flood management, and emergency preparedness. Cities can reduce both emissions and vulnerability through compact design, transit access, cooling, and flood risk reduction.

At the national level, priorities include clean electricity, industrial decarbonization, methane regulation, land-use policy, research and development, infrastructure planning, and carbon pricing or equivalent regulation. National policy is crucial because many mitigation and adaptation decisions depend on standards, finance, grids, markets, and institutions.

At the global level, priorities include coordinated emissions reductions, finance for mitigation and adaptation, technology transfer, methane and deforestation initiatives, and mechanisms for loss and damage. These are especially important for climate justice because vulnerability and historical responsibility are unevenly distributed [1].

14 Limitations and Data Provenance

This report is an evidence synthesis, not a new empirical climate attribution study. No empirical quantitative dataset was provided. Therefore, no independent statistical test of the hypothesis was performed. The conclusions are based on the supplied authoritative literature, especially IPCC AR6, NASA, NOAA, EPA, and the Royal Society/U.S. National Academy of Sciences.

The source set is also limited. The assignment requested use of sources such as WHO, WMO, UNEP, IEA, and World Bank where relevant, but those sources were not included in the final Sources block supplied for the paper. As a result, this report avoids citing claims that would require those sources, or discusses such topics only at the broad synthesis level supported by IPCC AR6. A fuller report should add the relevant Working Group II, Working Group III, WHO, WMO, UNEP, IEA, and World Bank sources.

Table 10: Practical action synthesis

Action area	Best-supported actions	Primary benefits	Key barriers
Energy	Renewables, efficiency, electrification	Emissions cuts, air quality, energy security	Grid, permitting, finance
Land	Avoid deforestation, restore ecosystems, protect soils	Carbon, biodiversity, water regulation	Land rights, permanence
Food	Reduce methane, improve fertilizer efficiency, reduce food waste	Emissions reduction and resilience	Incentives, access, behavior
Cities	Transit, cooling, flood planning, building codes	Health and resilience	Funding, governance, housing affordability
Industry	Efficiency, electrification, process innovation, selected carbon capture	Emissions reductions in hard-to-abate sectors	Cost, infrastructure, technology readiness
Finance and governance	Adaptation finance, mitigation finance, capacity building	Equity and implementation	Political will, institutional capacity

15 Conclusion

Global warming is the long-term increase in Earth’s average surface temperature and is one major component of broader climate change. The assessed authoritative literature supports the conclusion that human activities, principally greenhouse-gas emissions, have caused observed global warming [1]. The physical mechanism is the enhanced greenhouse effect, in which added greenhouse gases alter Earth’s radiative balance by trapping more outgoing infrared energy [3, 4, 5].

Observed consequences extend beyond temperature to sea level, ocean heat, land ice, Arctic sea ice, extremes, ecosystems, and human systems [1, 6]. Future risks increase with additional warming, and lower-emissions pathways reduce projected warming and associated risks relative to high-emissions pathways [1, 7]. The most defensible response is an integrated strategy of deep emissions reductions, methane abatement, clean energy, efficiency, electrification, sustainable land use, selected carbon capture for hard-to-abate sectors, demand-side change, and robust adaptation.

No independent empirical hypothesis test was performed in this report because no empirical dataset was provided. The assessed literature, especially IPCC AR6, supports the primary conclusion that anthropogenic greenhouse-gas emissions are the dominant cause of recent global warming and that lower-emissions pathways reduce future climate risks.

A Synthetic Workflow Demonstration

Warning: The values in this appendix are simulated and provide no empirical evidence about the real climate system. They are included only because a prior prototype workflow was supplied. The simulated data-generating process built in an anthropogenic forcing signal, so the regression results simply demonstrate that the workflow can recover an encoded relationship. These outputs are not used to support the scientific conclusions of this report.

B References

References

- [1] Intergovernmental Panel on Climate Change (IPCC). “Summary for Policymakers: Climate Change 2023.” 2023. <https://www.ipcc.ch/report/ar6/syr/summary-for-policymakers/>

Table 11: Synthetic workflow outputs; simulated values only

Quantity		Synthetic value	Interpretation
Anthropogenic tributable fraction	at-	0.997	Simulated workflow result; not an empirical attribution estimate
Synthetic 1850–2025	warming,	2.482°C	Simulated value; not an observed temperature estimate
Sea level slope		136.69 mm per 1°C	Simulated relationship; not a real-world estimate
Arctic sea ice slope		-1.64 million km ² per 1°C	Simulated relationship; not a real-world estimate
Low-emissions warming	2100	1.526°C	Simulated scenario; not an IPCC projection
Medium-emissions 2100 warming		2.703°C	Simulated scenario; not an IPCC projection
High-emissions warming	2100	4.174°C	Simulated scenario; not an IPCC projection

- [2] NASA Science. “What’s the Difference Between Climate Change and Global Warming?” n.d. <https://science.nasa.gov/climate-change/faq/whats-the-difference-between-climate-change-and-global-warming/>
- [3] Lindsey, Rebecca. “Climate Change: Atmospheric Carbon Dioxide.” NOAA Climate.gov, May 21, 2025. <https://www.climate.gov/news-features/understanding-climate/climate-change-atmospheric-carbon-dioxide>
- [4] U.S. Environmental Protection Agency (EPA). “Global Greenhouse Gas Overview.” December 30, 2025. <https://www.epa.gov/ghgemissions/global-greenhouse-gas-overview>
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- [6] Intergovernmental Panel on Climate Change (IPCC). “AR6 Synthesis Report: Climate Change 2023.” 2023. <https://www.ipcc.ch/report/sixth-assessment-report-cycle/2023/>
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